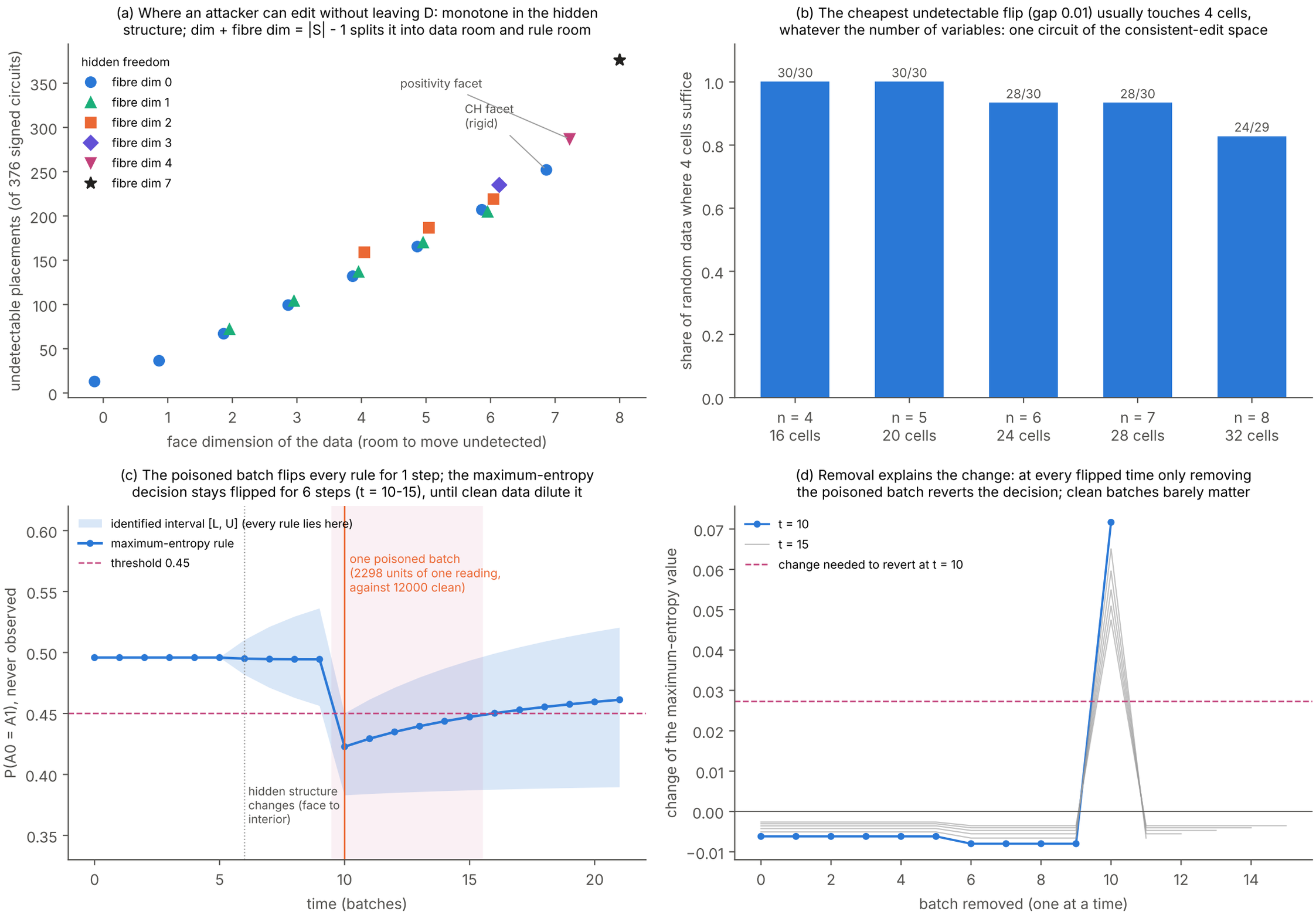
(a) Where an attacker can edit without leaving D: monotone in the hidden structure; dim + fibre dim = |S| - 1 splits it into data room and rule room
hidden freedom
fibre dim 0
fibre dim 1
fibre dim 2
fibre dim 3
fibre dim 4
fibre dim 7
positivity facet
CH facet (rigid)
undetectable placements (of 376 signed circuits)
face dimension of the data (room to move undetected)
(b) The cheapest undetectable flip (gap 0.01) usually touches 4 cells, whatever the number of variables: one circuit of the consistent-edit space
share of random data where 4 cells suffice
30/30
30/30
28/30
28/30
24/29
n = 4 16 cells
n = 5 20 cells
n = 6 24 cells
n = 7 28 cells
n = 8 32 cells
(c) The poisoned batch flips every rule for 1 step; the maximum-entropy decision stays flipped for 6 steps (t = 10-15), until clean data dilute it
identified interval [L, U] (every rule lies here)
maximum-entropy rule
threshold 0.45
one poisoned batch (2298 units of one reading, against 12000 clean)
hidden structure changes (face to interior)
P(A0 = A1), never observed
time (batches)
(d) Removal explains the change: at every flipped time only removing the poisoned batch reverts the decision; clean batches barely matter
t = 10
t = 15
change needed to revert at t = 10
change of the maximum-entropy value
batch removed (one at a time)